

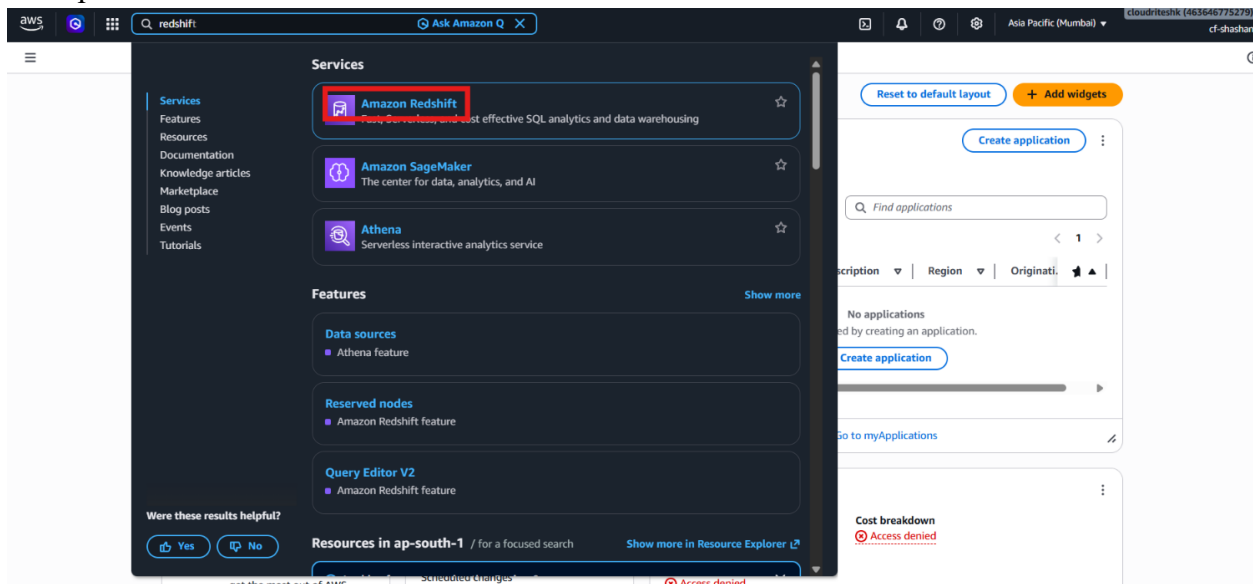
Getting Started with Amazon Redshift Serverless

To Begin with the Lab

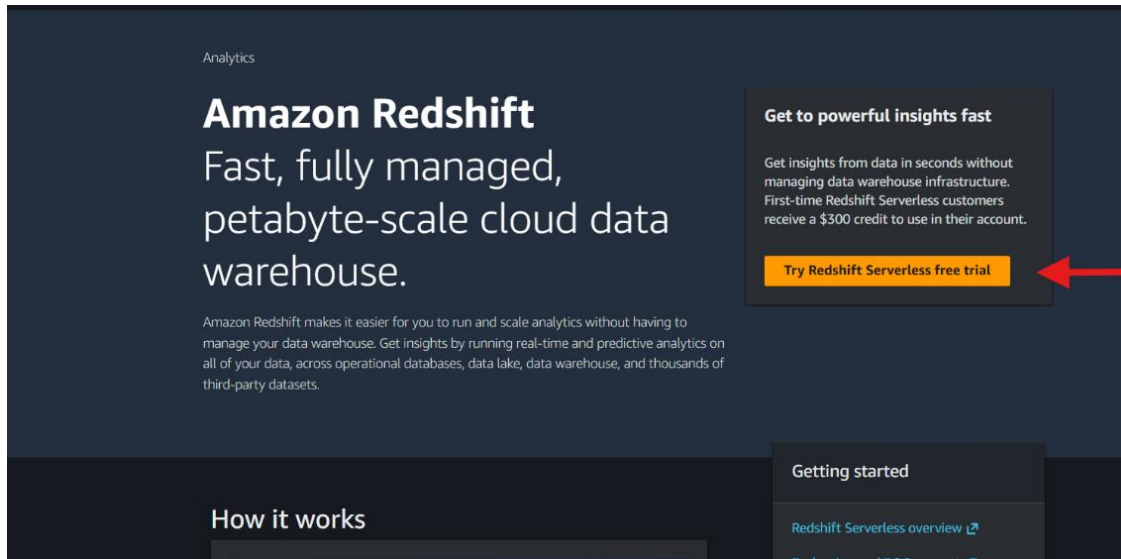
Summary of the Lab

In this lab, an **Amazon Redshift Serverless** data warehouse was created and configured using the default namespace and workgroup settings. An **IAM** role was generated to allow **Amazon Redshift** to access **Amazon S3** resources for loading and unloading data. The default networking configuration, including the VPC and security group, was used to simplify setup. After the serverless environment was provisioned, **Query Editor v2** was connected using federated credentials, and the Tickit sample dataset was loaded from **Amazon S3**. The setup was verified by executing the sample notebook queries and successfully viewing query results.

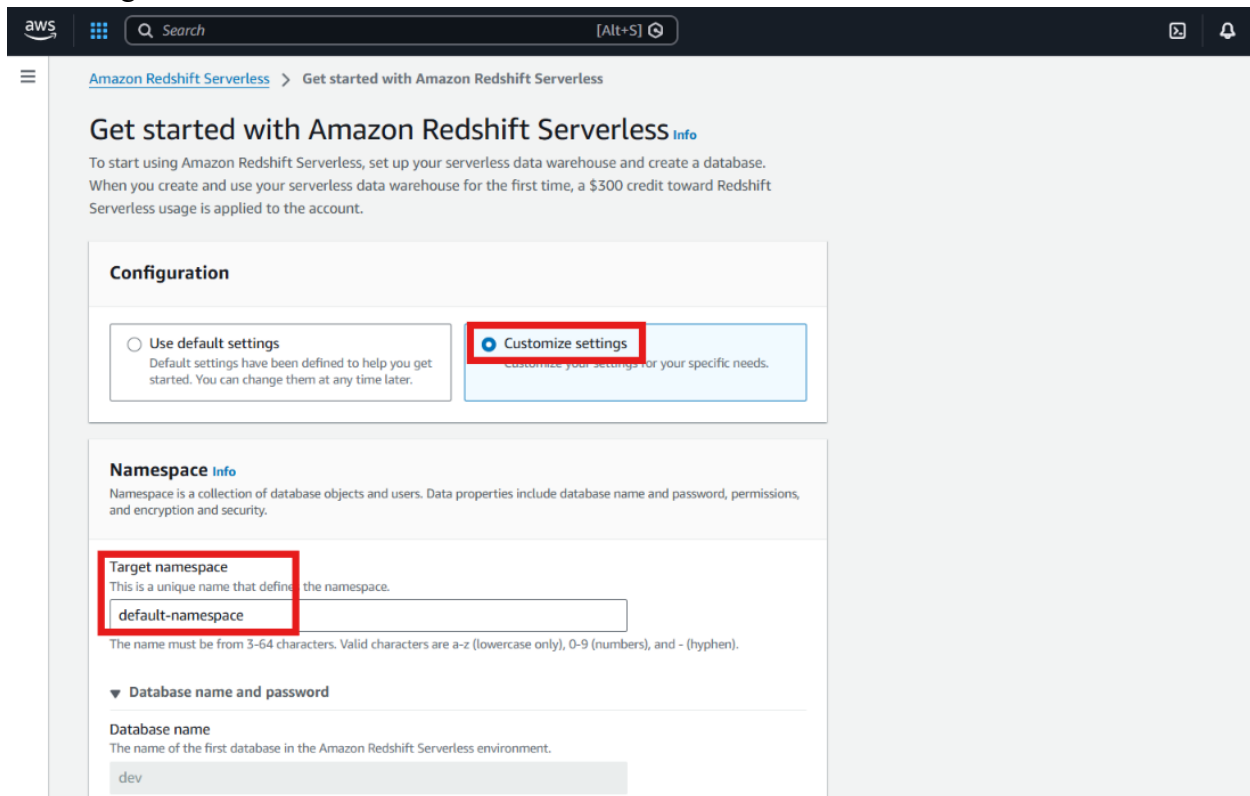
- Sign in to the AWS Management Console
- Search for **Amazon Redshift**
- Open **Amazon Redshift**



- Click **Try Redshift Serverless free trial**



- Click on **Customize settings** if you want to change name for namespace and other configurations.



- Navigate to **Workgroup** section and change name if you want to.
- Change **Base capacity** to lower value for lab purpose.

Workgroup info

Workgroup is a collection of compute resources from which an endpoint is created. Compute properties include network and security settings.

Workgroup name
This is a unique name that defines the workgroup.
default-workgroup

The name must be from 3-64 characters. Valid characters are a-z (lowercase only), 0-9 (numbers), and - (hyphen).

▼ **Capacity**

Set the base capacity used for processing your data warehouse workloads. Capacity is measured in Redshift processing units (RPU). Setting the RPU value higher increases capability and improves performance.

Base capacity
The default value is 128 RPUs. To change the base capacity, choose another RPU value.
8

From 4-8 RPUs the value increases in increments of 4 RPU. From 8-512 RPUs range must be in increments of 8 RPUs.

▼ **Track - new**

Track
The track type controls which release version is applied to the workgroup. [Learn more](#)

☒ **Current**
Use the latest workgroup version.

☐ **Trailing**
Use the version before the latest workgroup version.

▼ **Network and security**

IP address type [Info](#)

- Navigate to the **Permissions** section
- Click **Manage IAM Role** → **Create IAM Role** if not already created.

Database name and password

Database name: dev
Admin user credentials: Created based on IAM credentials

Permissions

❗ Associate an IAM role so that your serverless endpoint can LOAD and UNLOAD data. You can create an IAM role as the default for this configuration that has the [AmazonRedshiftAllCommandsFullAccess](#) policy attached. This policy includes permissions to run SQL commands to COPY, UNLOAD, and query data with Amazon Redshift Serverless. This policy also grants permissions to run SELECT statements for related services, such as Amazon S3, Amazon CloudWatch logs, Amazon SageMaker, and AWS Glue. You won't be able to run these SQL commands without an IAM role attached to your namespace.

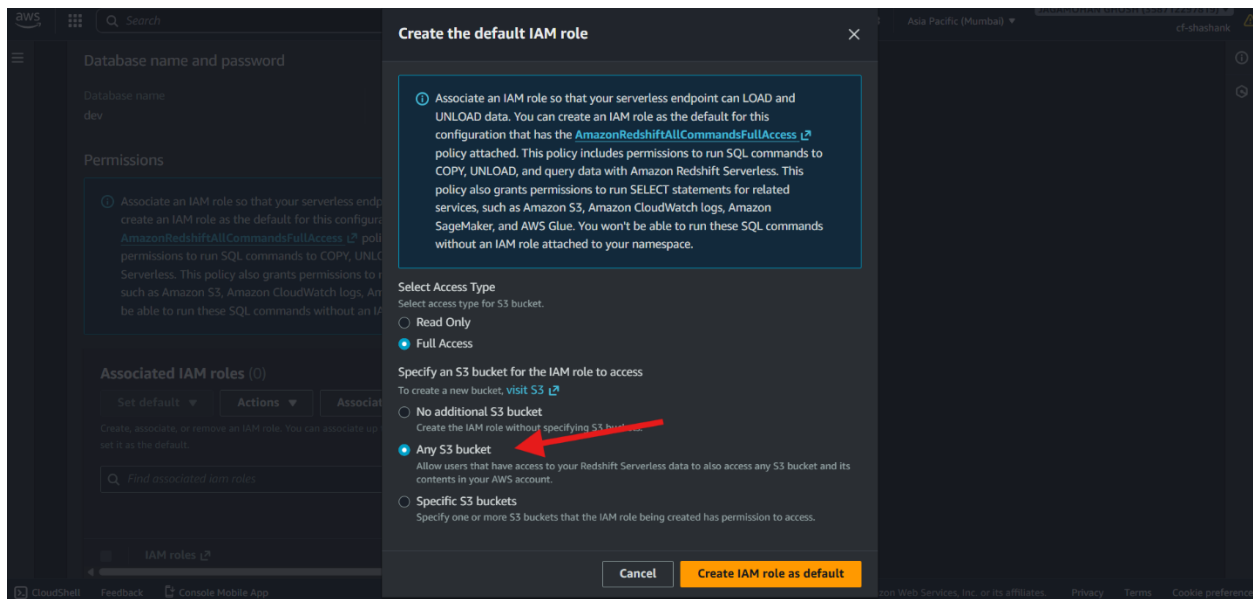
Associated IAM roles (0)

Set default ▼ Actions ▲ Associate IAM roles Create IAM role

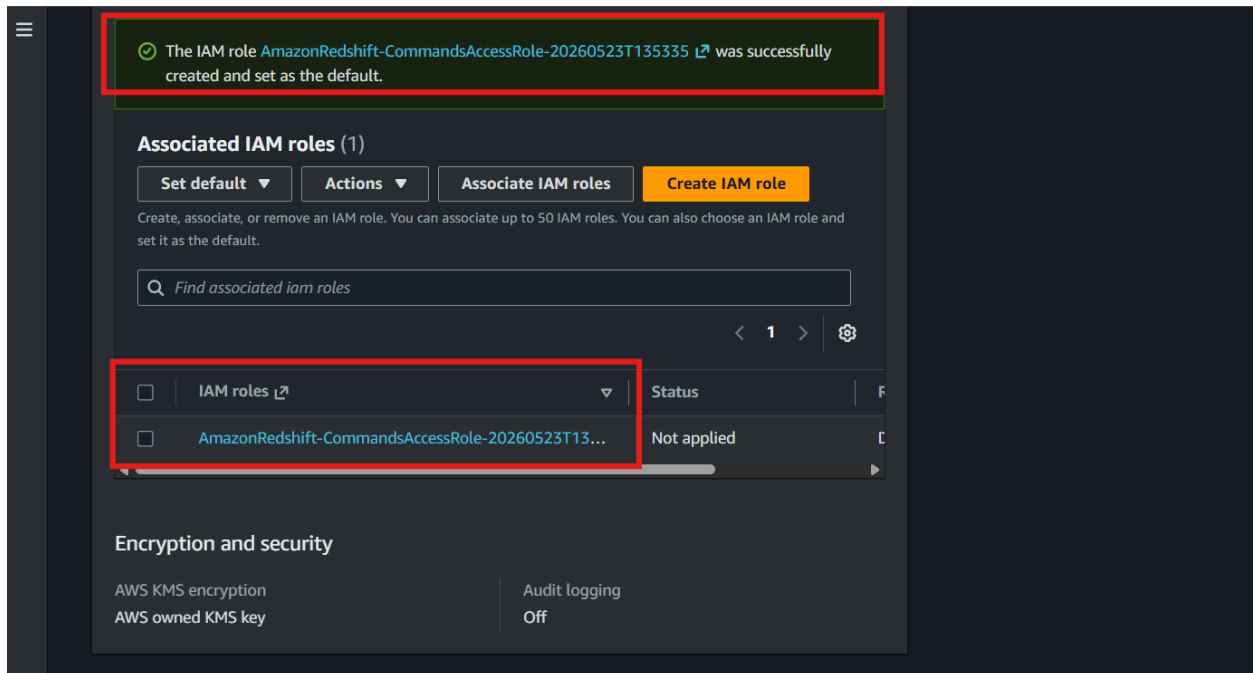
Create, associate, or remove IAM roles. Up to 50 IAM roles. You can also choose an IAM role and set it as the default.

Find associated IAM roles Remove IAM roles

- Select **Any S3 bucket** as the access type.
- Click **Create IAM role as default**.



- Wait for the default **IAM** role to be created.



- Review the automatically assigned VPC configuration.
- Review the automatically assigned security group configuration.
- Click **Save configuration**.

Base capacity [Info](#)
128

Track - new
Amazon Redshift periodically applies version updates to your workgroup based on your track setting. [Learn more](#)

Track
Current

Network and security

Virtual private cloud (VPC)
[vpc-0ae9dcd8b309b12cf8](#)

VPC security group
[sg-07258112f54495c0a](#)

SSL
Enable

Subnet
subnet-01c8167c1531504ef,
subnet-0dc3963e023360ccc,
subnet-0a200e2c0cec86709,

Enhanced VPC routing
Off

Autonomics configuration - new

Autonomics configuration
Not enabled with extra compute [Learn more](#)

Cancel **Save configuration**

- Wait for the serverless environment creation process to complete
- Click **Continue**
- Open the **Serverless dashboard**

Amazon Redshift Serverless

Serverless dashboard [Info](#) [Refresh](#) [Query data](#) [Go to provisioned clusters](#) [Create workgroup](#)

Namespace overview [Info](#) [Filter namespace](#)
All namespaces

Namespace data from your account

Total snapshots	Datashares in my account	Datashares requiring authorization	Datashares from other accounts	Datashares requiring association
0	0	0	0	0

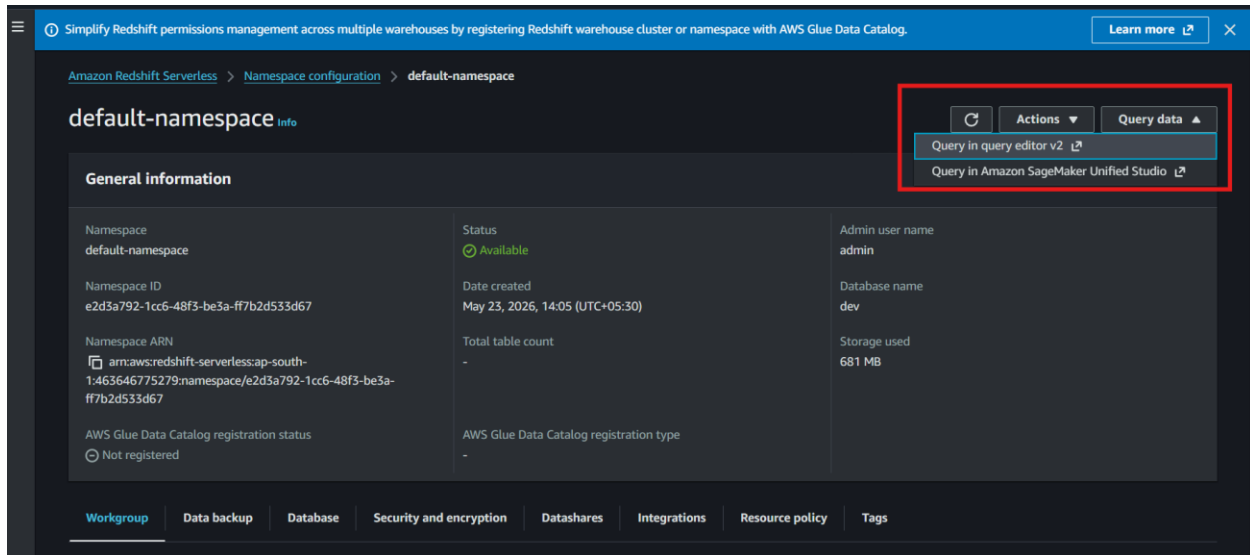
Namespaces / Workgroups [Info](#)

Namespace	Status	Workgroup	Status	Average query duration	Average number of
default-namespace	Available	default-workgroup	Available		

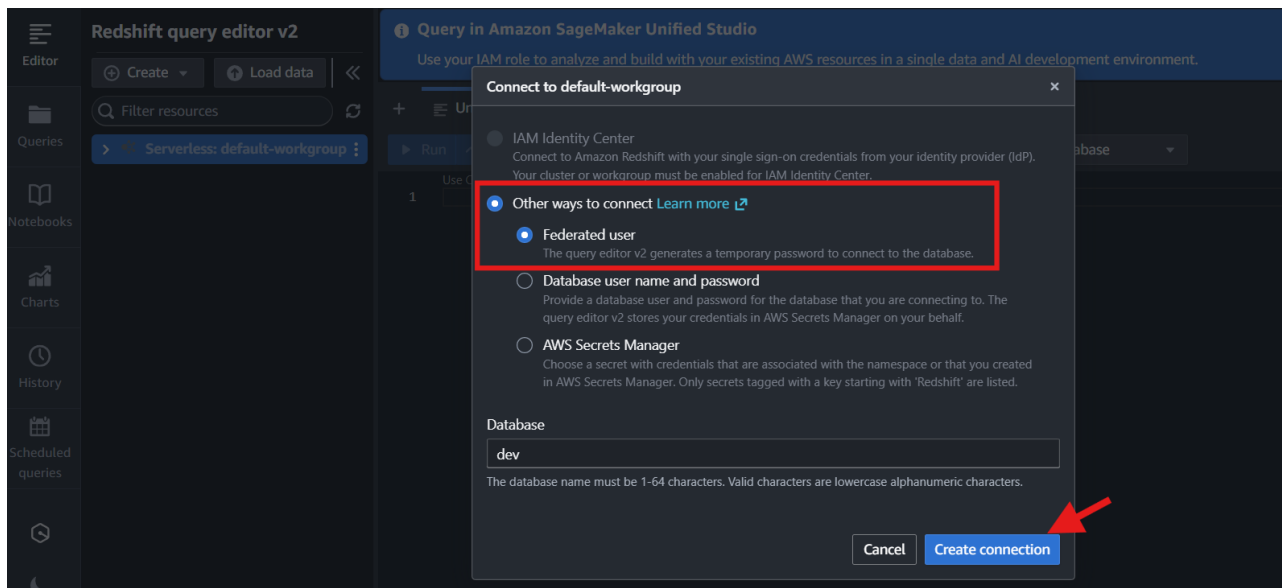
Total compute usage - new [Info](#)
Choose a workgroup Last hour
To visualise the costs of your total compute usage, go to [AWS Cost Explorer](#)

Total consumed RPU hours

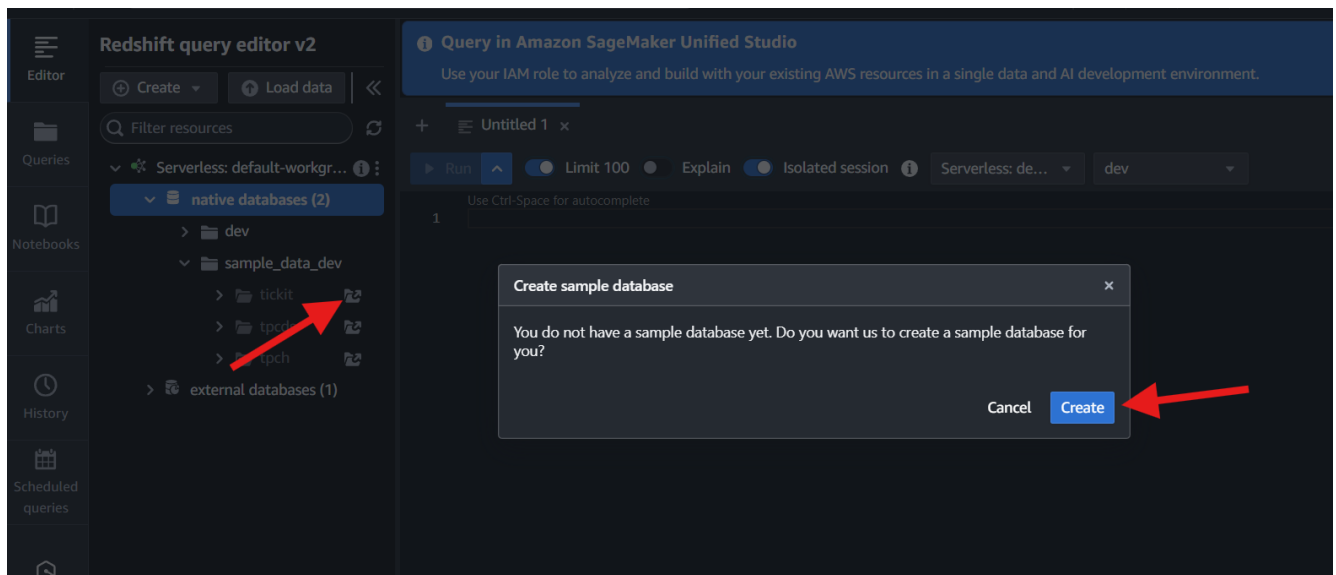
- Verify that the default namespace and workgroup are available
- Click the namespace name
- Click **Query data** → **Query in query editor v2**
- Open **Query Editor v2**



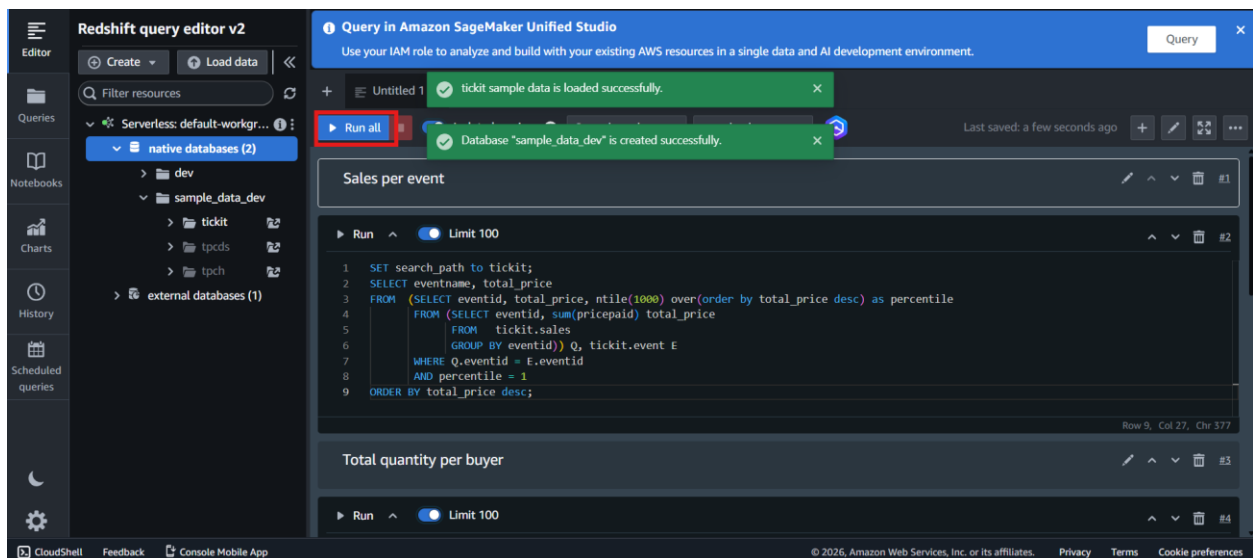
- Select the default workgroup connection
- Choose **Federated user** authentication
- Verify that the database name is set to **dev**
- Click **Create connection**



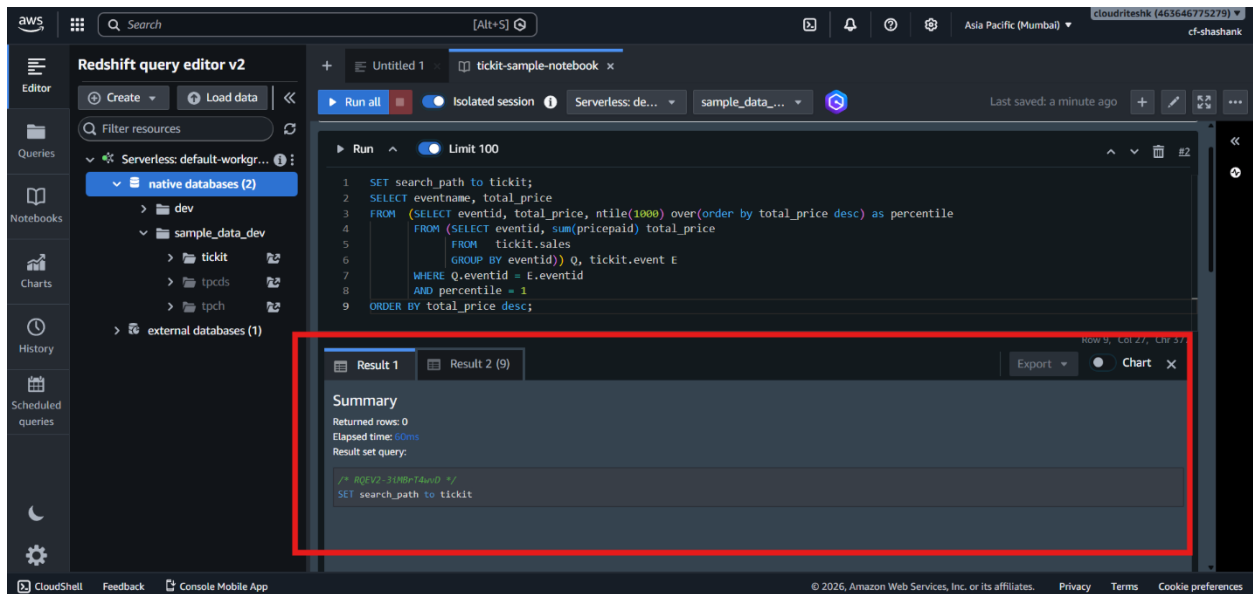
- Verify that the connection to the serverless workgroup is established
- Expand **Native databases**
- Expand **sample_data_dev**
- Review the available sample schemas
- Expand **ticket**
- Click **Open sample notebooks**
- Review the **Create sample database** prompt
- Click **Create**



- Wait for the Tickit sample dataset to load from **Amazon S3**
- Open the generated sample notebook
- Review the preloaded sample SQL queries
- Click **Run all**



- Wait for all notebook cells to execute



- Review the query results displayed below each notebook cell
- Verify that the Tickit sample dataset was loaded successfully
- Verify that **Query Editor v2** can execute queries against the **Amazon Redshift Serverless** database successfully